

BINARY SEARCH PATTERNS CHEAT SHEET

1. GLOBAL / STRUCTURE-BASED BINARY SEARCH Used when array has global ordering or structure.

Template: while(left < right): mid = ... if condition with boundary: left = mid + 1 else: right = mid

Examples: - Search Insert Position - Find Minimum in Rotated Sorted Array - Search in Rotated Sorted Array - First & Last Occurrence - Lower Bound / Upper Bound

Key Signal: sorted, rotated, first, last, min, max

2. LOCAL / SLOPE-BASED BINARY SEARCH Used when decision depends on local neighbors.

Template: while(left < right): mid = ... if nums[mid] > nums[mid+1]: right = mid else: left = mid + 1

Examples: - Find Peak Element - Peak Index in Mountain Array - Local Minima - Bitonic Array Peak

Key Signal: peak, mountain, local, greater than neighbors

DECISION RULE: If comparison is with boundary -> GLOBAL If comparison is with neighbor -> LOCAL